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SAYLANI HACKATHON · CHALLENGE BRIEF

HCM Lite

AI-assisted Human Capital Management

for Admin, HR and Employees



Admin · HR · Employee · 9 modules · AI built in

24 October 2026

Zaitoon Ashraf IT Park

Submit on lms.saylaniit.com

HCM Lite — Challenge Brief

Build an AI-assisted HR system for Admin, HR and Employees

1 The challenge

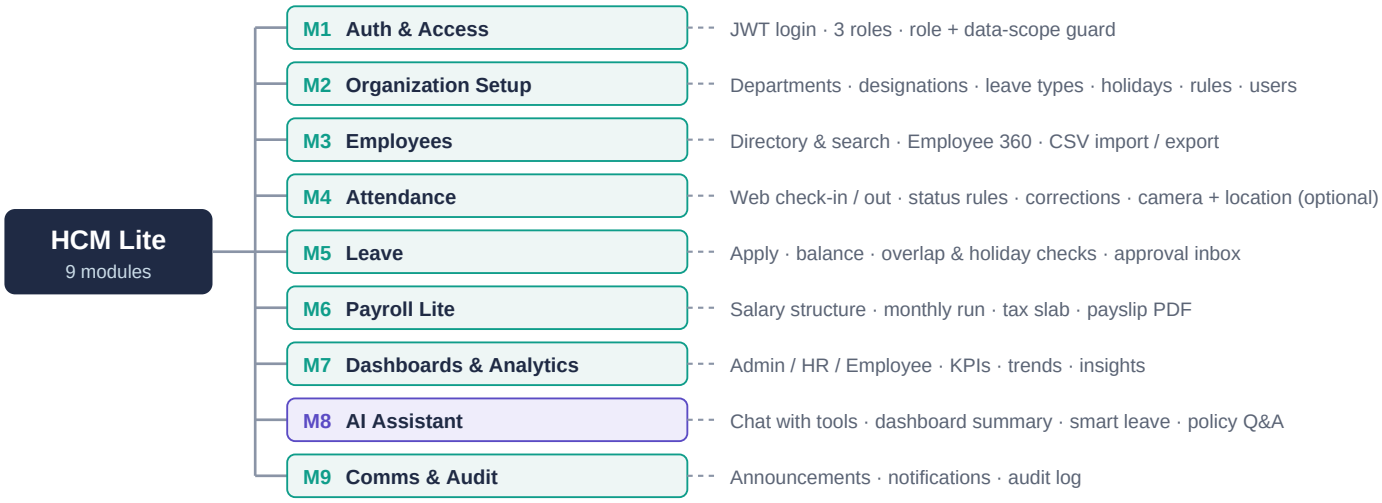
At the Saylani Hackathon, build **HCM Lite**: a web-based Human Capital Management system for the daily work of an HR team, with an **AI assistant** that answers questions from real data. There is **no pre-loaded or dummy data**: the system starts empty, Admin sets up the company, HR registers employees, and every number comes from what people do in the app.

Team	Date & venue	Work	Data	AI
4 people (2 backend, 2 frontend)	24 Oct 2026, Zaitoon Ashraf IT Park	25 functionalities across 9 modules	Starts empty; created by Admin, HR and employees in the app	Any hosted LLM API with tool calling (free tier is enough)

Teams must ship

- A web app with **three roles** (Admin, HR, Employee), each with its own menu, data scope and dashboard.
- **Core HR flows end to end**: employees, web check-in/out, leave with approval, monthly payroll with payslips.
- **Analytics** on the data entered in the app and an **AI assistant**, dashboard summaries and a smart leave form.
- A **web app is required**. A mobile app with camera attendance is optional and earns bonus points (Section 5).
- A 5-minute live demo and a README a judge can run in under 10 minutes. Any language, framework or database is allowed.

Figure 1 — Module tree



HCM Lite — Challenge Brief

Build an AI-assisted HR system for Admin, HR and Employees

2 Access levels: who can do what

Three roles. Access is checked **on the server** for every request; hiding a button is not enough. The AI assistant always works with the same access as the person using it.

Admin — company level	HR — people operations	Employee — self-service
• Company setup: name, office hours, grace time, weekend, **office location and radius** for camera attendance • Departments, designations, leave types and yearly quotas, public holidays • Create, deactivate and reset logins for HR and employees • View every employee, attendance, leave and payroll record • **Finalize** the monthly payroll prepared by HR (only Admin can lock it) • Company dashboard: headcount, attrition, payroll cost, average salary • Audit log of every important action; post announcements • AI assistant on all company data	• **Register employees** (this creates their login); edit details; optional CSV import and export • Set salary structures (basic, allowances) • Daily attendance board; approve or reject attendance corrections • Approve or reject leave requests with a comment • Create and review the monthly payroll draft • HR dashboard and insights (late-comers, absenteeism, leave use) • AI assistant and dashboard summaries on HR data • Post announcements	• Check in and check out (web; camera + location on mobile, optional) • Own attendance calendar; request a correction with reason • Apply for or cancel leave, see balance; smart leave in plain words • Download own payslips • View own profile; edit phone and address • Personal dashboard: hours, late count, leave left, last payslip • AI assistant on own data and company policy • Read announcements and notifications
Cannot	**Cannot**	**Cannot**
• Edit a payroll run after it is finalized • Delete or change audit log entries	• Change company settings, rules or office location • Create Admin users or finalize payroll • View the audit log	• See any other employee's data or salary • Approve anything, or edit own salary, role or attendance directly

First run: a fresh install asks for the company name and the Admin account (or reads ADMIN_EMAIL / ADMIN_PASSWORD from the env file). Admin then creates HR logins; HR registering an employee creates that employee's login with a temporary password that must be changed on first sign-in.

3 Business rules

Module	Rules that decide whether it works
M1 Auth	JWT login, bcrypt passwords. Every API checks role **and** data scope: an Employee only reads rows with their own employee_id.
M2 Setup	Office start 09:00, grace 15 min, half day under 5 h, weekend Sat–Sun, office location + radius (e.g. 150 m). Stored as settings, applied to new records only.
M3 Employees	HR registers each employee with job details and salary; this creates the employee's login (temporary password, change on first sign-in). Server-side search, filters, paging. Optional CSV import has a **dry run** returning row + field + reason; checks required fields, unique code/email, valid email, known dept, ISO date, salary > 0.
M4 Attendance	Server stamps the time. One record per employee per day (second check-in = 409). Late after 09:15; half_day under 5 h; absent if no record and no approved leave; missing_checkout if no check-out.
M5 Leave	Working days only (skip weekends, holidays). Quotas annual 14, sick 8, casual 10; unpaid unlimited. Overlap or over-balance = 422.
M6 Payroll	Gross = 1.5 × basic (HRA 40%, medical 10%). Per day = gross ÷ working days; deduct absent + unpaid + 0.5 × half days. Prorate joiners/leavers. Tax slab on gross × 12 (illustrative): 0 to 600k; 5% to 1.2M; 30k + 15% to 2.4M; 210k + 25% to 3.6M; 510k + 35% above. Admin finalizes; finalized runs are locked.
M7 Dashboards	All numbers computed on the server (database aggregation) and filterable by date range and department (see Section 7).
M9 Comms & Audit	Announcements on all dashboards, in-app notifications, audit log with actor, action, entity, before/after JSON.

4 AI Assistant (M8)

Use any hosted LLM API that supports **tool calling**. The AI explains and assists; the database is the only source of truth.

Feature	Who	How it works
Ask the assistant	All roles	Chat panel → POST /ai/chat. The model picks a tool, the backend runs it with the user's scope, the model writes the answer. Tools: get_headcount, get_attendance_summary, get_late_comers, get_leave_summary, get_my_leave_balance, get_payroll_summary, search_employees.
Dashboard summary	Admin, HR	'Summarize' sends the KPI JSON for the current filters; returns 3–5 insight bullets using only those numbers.
Smart leave	Employee	"Need next Monday and Tuesday off for a wedding" → JSON {type, start, end, reason} → pre-filled form. The user confirms.
Policy Q&A;	All roles	Rules from Section 3 sit in the system prompt; the model answers only from them or says it does not know.

Guardrails

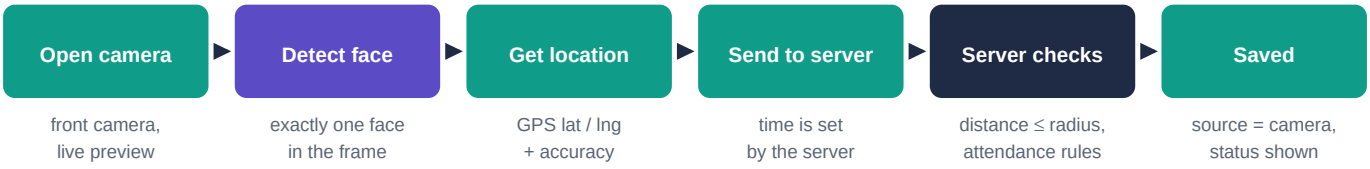
- API key only in a backend env variable (`AI_API_KEY`); the browser never calls the AI provider.
- The model never queries the database directly. Tools reuse the same services and permission checks as the REST API.
- The AI never saves data on its own. JSON output is validated; on failure retry once, then show the normal form.
- 15 s timeout and a friendly fallback, about 20 requests per user per hour, every call logged in `ai_logs`. Answers carry an 'AI-generated' label.

Live test questions judges will ask

As	Question	Good answer
HR	How many active employees are in {department}? · Who came late most often this month? · What is this month's attendance rate?	Same numbers as the dashboard
Employee	How many sick leaves do I have left this year?	Own balance after approved leaves
Employee	What is {colleague}'s salary?	Politely refuses
Employee	I need next Monday and Tuesday off for my sister's wedding	Pre-filled casual leave, correct dates

5 Optional: mobile app and camera attendance

The **web app is required**. A mobile app is optional and earns bonus points. Its key feature: mark attendance with the **camera**, accepted only when a face is detected and the phone is **inside the company location** set by Admin. The same camera check can also run in the web app on a laptop or phone browser.



- Reject the check-in if no face or more than one face is found, if the location is outside the radius, or if GPS accuracy is worse than 100 m.
- The **server** calculates the distance to the office (Haversine formula) and decides; the phone only sends data. Admin sets office lat / lng and radius.
- Store the result, not a full photo, unless the company asks for one; tell users what is captured. A failed check falls back to an HR correction.
- **Extra bonus:** match the face to the employee's enrolled face (face descriptors) and a simple liveness check such as a blink.

Free libraries

Platform	Library	What it gives you
Web (MERN / React)	**face-api.js** (maintained fork: @vladmandic/face-api)	Face detection, landmarks and face descriptors for matching, in the browser
	MediaPipe Face Detector (@mediapipe/tasks-vision)	Fast real-time face detection from the webcam
	TensorFlow.js face-detection (@tensorflow-models/face-detection)	Lightweight face detection model; works with browser Geolocation API for location
React Native	**react-native-vision-camera** + react-native-vision-camera-face-detector	Camera with real-time on-device face detection
	@react-native-ml-kit/face-detection	Detect faces in a captured photo on the device
	expo-camera + **expo-location**	Easiest Expo setup for taking the photo and reading GPS
Server (Node)	**geolib**	isPointWithinRadius() for the office geofence check

All listed libraries are free and open source. Expo needs a development build for the ML Kit based face libraries.

6 Project overview

How the pieces fit together. Teams are free to choose any database (MongoDB, Firebase, PostgreSQL, MySQL or another) and to model the data as collections or tables however suits their stack.

Figure 2 — System overview

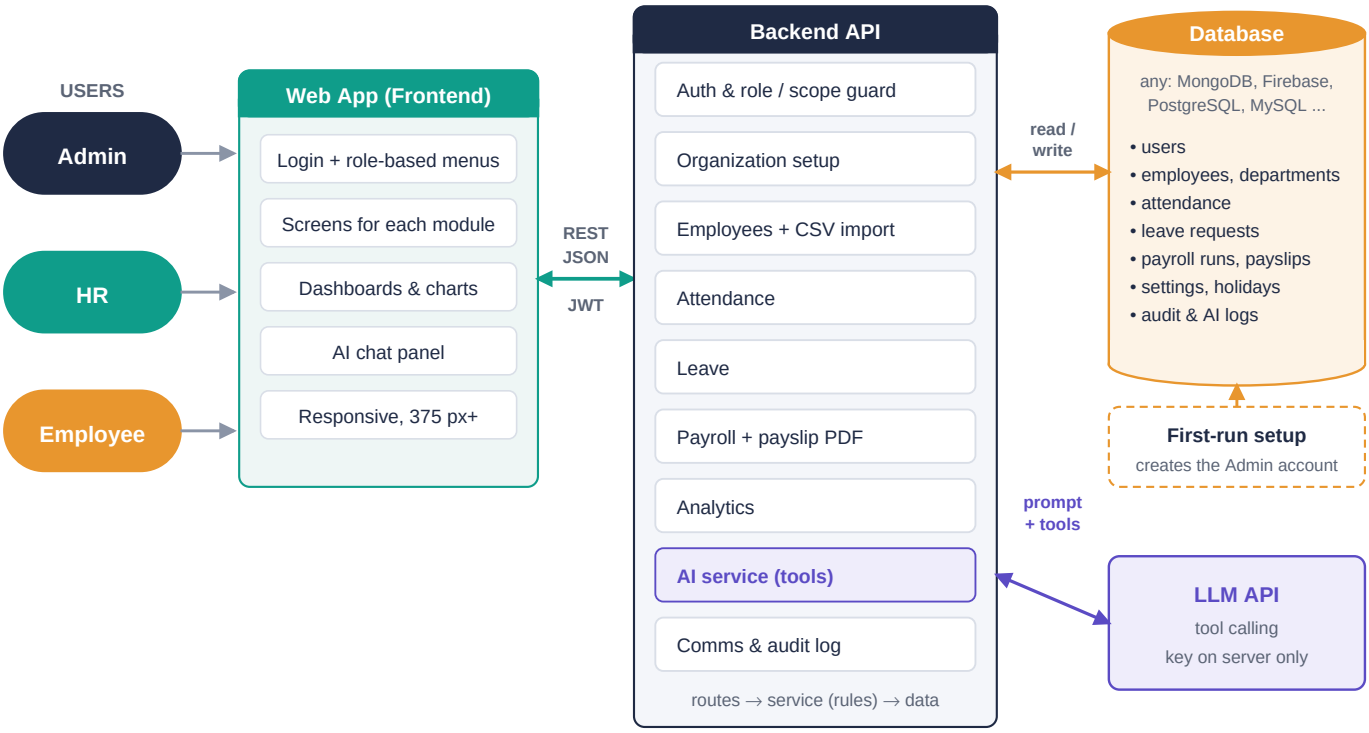
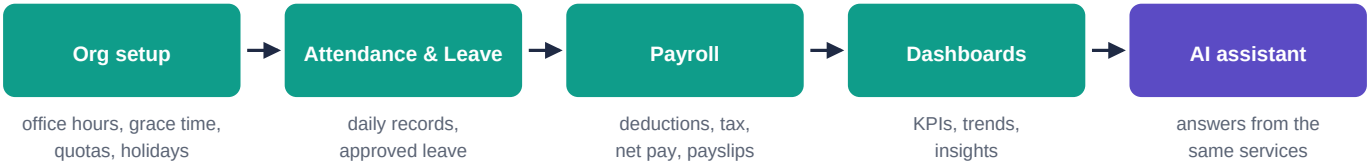
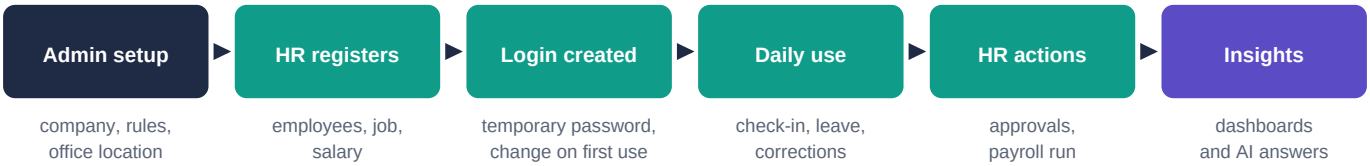


Figure 3 — How data flows between modules



Rules set in Org setup decide attendance and leave status; attendance and unpaid leave decide payroll deductions; dashboards aggregate all of it; the AI assistant calls the same services, so its answers always match the dashboards.

Figure 4 — How data enters the system (no dummy data)



Judges start from a fresh install during judging and create the company, a few employees and their activity themselves, then check that dashboards and AI answers match those records.

7 Metrics and dashboards

Metric	Formula
Headcount / attrition	Active + probation on end date / exits in period ÷ average headcount × 100
Attendance rate	(present + late + missing_checkout + 0.5 × half_day) ÷ (working-day records – on_leave)
Late % / absenteeism	late ÷ (present + late) / absent days ÷ expected working days, per department
Leave utilization	Approved days ÷ quota, per leave type and department
Payroll cost / avg salary	Sum of gross and net by month and department / mean gross by department and designation

HR dashboard	Admin dashboard	Employee dashboard
KPIs: headcount, present, on leave, late today, pending approvals · attendance trend · absenteeism by dept · leave by type · top 5 late-comers · AI summary	KPIs: headcount, attrition YTD, payroll cost, avg salary · headcount by dept · payroll cost by dept · gender and tenure · audit feed · AI summary	Check-in widget · hours this week vs 40 h · late count · leave balance · last payslip · announcements · ask my assistant

8 Functionality checklist

25 functionalities and who is allowed to use each one. **Manage** = create and change, **All** = see every record, **Own** = only their own records, **—** = no access.

#	Functionality	Admin	HR	Employee
01	First-run setup: company profile and the Admin account	Manage	—	—
02	Login, logout, change password; role-based menus and dashboards	Own	Own	Own
03	Company settings: office hours, grace time, weekend, office location and radius	Manage	View	—
04	Departments, designations, leave types and quotas, public holidays	Manage	View	—
05	Create HR logins; deactivate or reset any login	Manage	—	—
06	Register an employee (creates their login) and edit details	View	Manage	Own phone & address
07	Employee directory with search and filters; Employee 360 profile	All	All	Own
08	CSV import with dry-run error report; CSV export	—	Manage	—
09	Check in and check out on the web	—	Own	Own
10	Attendance calendar and daily attendance board	All	All	Own
11	Attendance correction: request and approve	View	Approve	Request
12	Apply for or cancel leave; see balance and history	—	Own	Own
13	Approve or reject leave with a comment	View	Approve	—
14	Salary structure per employee	View	Manage	—
15	Monthly payroll: create and review the draft	View	Manage	—
16	Finalize payroll (locks the month)	Manage	—	—
17	Payslips with PDF download	All	All	Own
18	Dashboards, KPIs, trends and insight lists with filters	Company	HR	Personal
19	AI assistant chat answering from real data	All data	HR data	Own data
20	AI dashboard summary	Yes	Yes	—
21	Smart leave: type a sentence, get a pre-filled leave form	—	Own	Own
22	Policy Q&A on attendance, leave and payroll rules	Yes	Yes	Yes
23	Announcements	Post	Post	Read
24	In-app notifications (leave decisions, payslips)	Own	Own	Own
25	Audit log of create, update, approve and finalize actions	View	—	—

Bonus (+10 points): camera attendance with face detection + location (mobile or web; Employee uses it, Admin sets the location) · attrition risk score · absence anomaly flags · late penalty rule (3 lates = 0.5 day) · AI understands Roman Urdu · dark mode / PWA.

9 Demo and judging

5-minute demo script

- Start from a fresh install: **Admin** sets up the company, office hours and location, a department and leave types, then creates an HR login.
- **HR** registers two or three employees; one of them signs in, changes the password and checks in.
- **Employee** types a leave request in plain words and confirms the pre-filled form; **HR** approves it.
- **HR** opens the dashboard, clicks *Summarize* and asks the assistant who is on leave; runs payroll for the month; **Admin** finalizes it.
- **Employee** downloads the payslip and asks for a colleague's salary (refused); **Admin** shows the audit log.

Scoring (100 points + 10 bonus)

Criteria	Pts	What judges look at
Core HR flows	25	Login per role, employees, check-in, leave approval, payroll run end to end
Data & analytics	20	Dashboards match the records created during judging; filters work; insights are useful
AI features	15	Correct live answers, role scoping, smart leave, summaries, graceful failure
Code quality	15	Module structure, rules in services, validation, tests
UI / UX	10	Clarity, consistency, loading / empty / error states, responsive
Security & RBAC	10	Server-side scope checks, hashed passwords, AI key on server, audit log
Demo & docs	5	Clear 5-minute story; working hosted link; README a judge can follow

Submission

Submit on the Saylani LMS: https://lms.saylaniit.com/
Every student logs in to their own portal and shares **two links**: the **Git repository** and the **hosted (live) project**.

- **Repository link:** public, or shared with the judges. README with the stack, how to run it, how to create the first Admin, how to set the AI key, and what is done and not done.
- **Hosted project link:** the live web app opens in a browser and starts from first-run setup, so judges can create the company themselves. Any free hosting is fine (frontend and backend).
- Commit `.env.example` only; never the real AI key or database password. Keep secrets in the hosting provider's environment settings.
- Mobile app (optional): add an APK or Expo link in the README.

Best of Luck!

Build the core first, then make the data and the AI talk. We can't wait to see what you create.

Saylani Hackathon · 24 October 2026 · Zaitoon Ashraf IT Park